primary objectives might be to reduce water and electricity consumption. The following example works through these objectives based on the process defined earlier, starting with identifying opportunities together with the key data needed to enable better decisions, as shown in table 2.3.

Table 2.3 Start by defining your objectives in terms of the opportunities for improvement your digital twin will offer and what data is required.

Opportunity	Annual cost	Current decision	Better decision	Key data needed
More effective use of electricity produced by solar PV	$2,160	Run appliances when convenient	Schedule high-power loads during optimal solar production	Real-time electricity usage, solar PV output, appliance scheduling
Reduce excessive water usage by over-irrigation	$456	Water based on a fixed schedule	Irrigate based on soil moisture & weather forecast	Soil moisture sensors, rainfall forecast, and irrigation flow rates
More efficient cooling/heating decisions	$1,800	Adjust temperature for immediate comfort	Optimize timing with solar production and occupancy	Indoor/outdoor temperature, occupancy detection, HVAC energy usage
Mitigate risk of fire & flood	$50,000 potential	Only smoke detectors	Leak detection	Moisture, water flow measurement

These opportunities then lead to defining my metrics for success. For my home digital twin, these are the modest metrics listed in table 2.4. These metrics are the lagging indicators that I will measure to determine if I have met my objectives. Depending on the scope of your proposed digital twin, you may have significantly more metrics—the key is to understand what they are, the baseline measure, and your proposed target.